UP Board 2020

Grade 12

Physics

Time: 3 Hours 15 Minutes **Max. Marks: 70**

Note: First 15 minutes are allotted for the candidates to read the question paper.

Instruction :

i) All questions are compulsory.

ii) This question paper has 5 sections: Section A, Section B, Section C, Section D and Section E.

iii) Section A is of multiple choice type and each question carries 1 mark.

iv) Section B is of very short answer type and each question carries 1 mark.

v) Section C is of short answer I type and carries 2 marks each.

vi) Section D is of short answer II type and carries 3 marks each.

vii) Section E is of long answer type. Each question carries 5 marks. All four questions of this section have been given internal choice. You have to do only one question from the choice given in the question.

SECTION - A

1 a) The sides of a cube of the same material are l and 2l respectively. The ratio of their resistances will be

i) 1 : 1

ii) 1 : 2

iii) 2: 1

iv) 4: 1. **1**

b) The resistance of the required shunt in order to pass 100% of the main current in a galvanometer of 99Ω resistance should be

i) 11 Ω

ii) 10 Ω

iii) 9 Ω

iv) 9.9 Ω **1**

c) If wavelengths of γ-rays, X-rays and micro-waves in vacuum are λ 1, λ 2 and λ 3 respectively then

i) $\lambda\ 1 = \lambda\ 2 = \lambda\ 3$

ii) $\lambda\ 1 < \lambda\ 2 > \lambda\ 3$

iii) $\lambda\ 1 > \lambda\ 2 < \lambda.3$

iv) $\lambda\ 1 < \lambda\ 2 < \lambda\ 3$ **1**

d) When a ray of light enters in a medium of refractive index n, the angle of refraction is half of the angle of incidence. Value of the angle of refraction will be

i) $2\cos^{-1}(n)$

ii) $\cos^{-1}\left(\frac{n}{2}\right)$

iii) $2\sin^{-1}\left(\frac{n}{2}\right)$

iv) $2\cos^{-1}\left(\frac{n}{2}\right)$ **1**

e) Photons of 6 eV energy are incident on a metal surface, then maximum kinetic energy of the emitted photoelectrons is 5 eV. Stopping potential should be

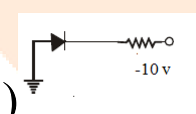
i) 5 volt

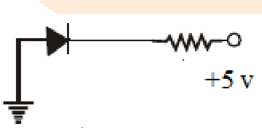
ii) 1 volt

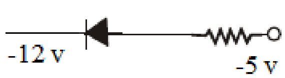
iii) 6 volt

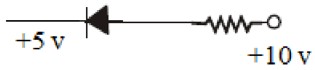
iv) 11 volt. **1**

f) Which of the following p-n junction diodes is reverse-biased ?

i) 

ii) 

iii) 

iv) 

SECTION – B

2. a) Electromotive force of a cell is 2 volt. What is the meaning of this statement ? **1**

b) Magnetic susceptibility of a substance is 0-9853. Find out the relative permeability of the substance. **1**

c) Define 1 henry of self inductance. **1**

d) The amplitude of electric field is 3.0×10^{-4} VM $^{-1}$ of a plane electromagnetic wave. Find out the amplitude of its magnetic field. **1**

e) A nucleus $z^{X^{A}}$ emits one a-particle and one β-particle respectively. What will be the new nucleus after the emission ? **1**

f) What is the principle of a photo-diode ? **1**

SECTION – C

3. a) A cell of e.m.f. 10 V and internal resistance 3Ω is connected to a resistor in series. If the current in the circuit is 0.5 A, then find out the resistance of the resistor and terminal voltage across the cell. **2**

b) An object is kept in front of a concave mirror of 15 cm focal length. The image formed is real and three times the size of the object. Find the distance of the object from the mirror. **2**

c) Obtain the ratio of the accelerating potentials required for a proton and an a-particle to have the same de Broglie wavelength associated with them. **2**

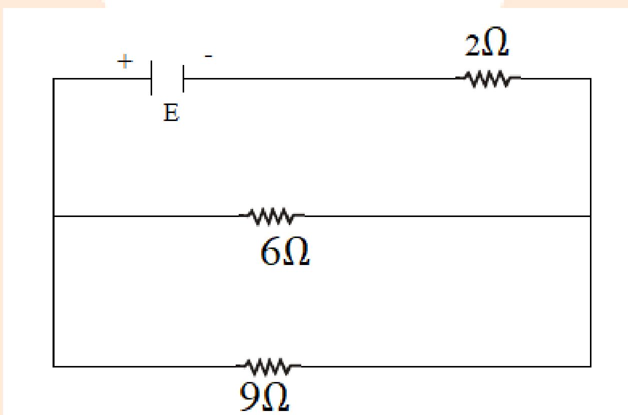
d) How is AND gate obtained by NOR gate ? **2**

OR

How is OR gate obtained by NOR gate ? **2**

SECTION – D

4. a) If dissipated heat in 9 Ω resistor is 36 watt, then find the heat dissipated in 6 Ω resistor and potential difference across the ends of 2 Ω resistor. **3**



b) Obtain the formula for the angular width of central maxima of the diffraction pattern of light by a single slit by drawing a diagram. Draw also the intensity distribution diagram of diffraction pattern. **3**

c) Write down Einstein's photoelectric equation. When

wavelength of the incident light on a metal surface is λ_1, then stopping potential is V_1 and for wavelength λ_2, stopping potential is V_2. Obtain the formula for the Planck's constant.

3

d) What is the nuclear fusion process ? Give an example of

nuclear fusion in nature.

A star converts four helium nuclei into one oxygen nucleus.

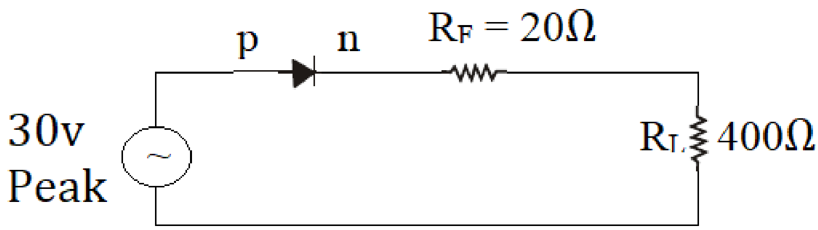
If masses of helium and oxygen nuclei are 4.0015 amu and

15.995 amu respectively , then find the energy released per

oxygen nucleus. **3**

e) An ac voltage of peak value 30 V, is connected in series with

a germanium diode and a load resistor of 400 Ω. The forward resistance of the diode is 20 Ω and the barrier voltage is 0.3 V. Find the peak current through the diode and peak voltage across the load. What will happen to these values, if the diode is assumed to be ideal ? **3**



5.

a) Obtain the formula for the intensity of electric field near a long charged wire with the help of Gauss' law. Linear charge density of wire is λ Cm-1 . **3**

b) Write down the vector form of Lorentz force acting on a

moving charge in a uniform magnetic field. Two particles A and B having same charge after being accelerated by the same accelerating potential, enter perpendicular in a uniform magnetic field, so that they R_1 and R_2 respectively. Find out the ratio of the masses of A and B in terms of R_1 and R_2 **3**

c) The length of a wire becomes three times by stretching it. The

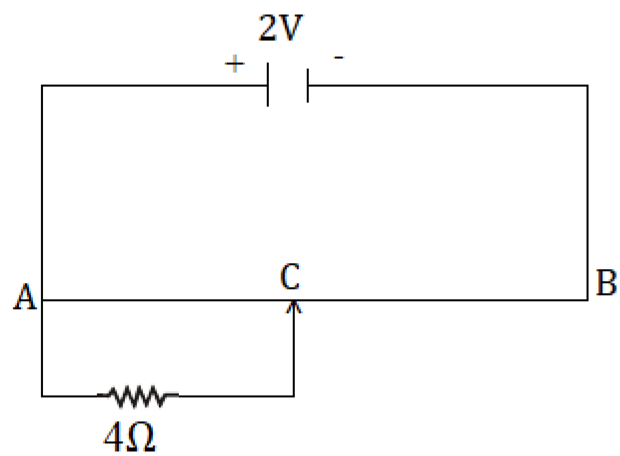
wire is connected with a source of constant voltage. What would be the effect on (i) resistance,

(ii) resistivity of the material and

(iii) drift speed of the free electrons of the wire ? **3**

OR

What is the principle of a potentiometer ? The resistance of a potentiometer wire is 8 Ω. Find the potential difference across the 4Ω resistor, when the sliding jockey is just in contact at the mid-point C of the potentiometer wire : **3**



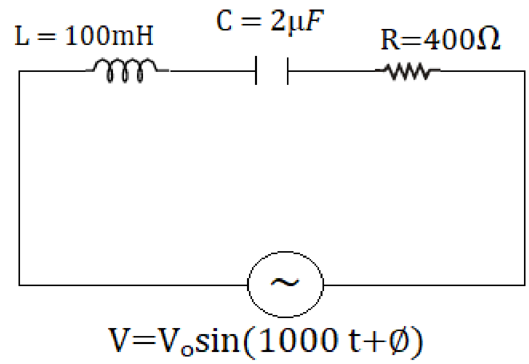
d) Draw a labelled ray diagram of an astronomical telescope, when final image is formed at the least distance of the distinct vision. Which of the following convex lenses are to be used for making a telescope having maximum magnifying power ? Also find out the magnifying power of this telescope :

$f_1 = 20$ cm, $f_2 = 50$ cm, $f_3 = 1$ m, $f_4 = 5$ cm. **3**

e) Explain the working process of full wave rectification of (p-n) junction diode. **3**

SECTION – E

6. Find (i) the value of the phase difference between the current and the voltage in the given series LCR circuit, (ii) if a capacitor C_1 is connected in parallel with the capacitor C in order to make the power factor of the circuit unity (1), then find out the value of the capacitance of C_1 .



5

OR

Define coefficient of mutual induction. A circular coil X of radius 0.02 m and 100 turns is placed coaxially and concentric with another coil Y of radius 0.2 m, having 1000 turns. Find mutual inductance of the coils, when current in the coil Y changes from 7A to 5A in 4 x 10^{-2} s. What is the e.m.f. induced in the coil X ? **5**

7. Derive the formula for the refraction at a convex spherical surface. **5**

OR

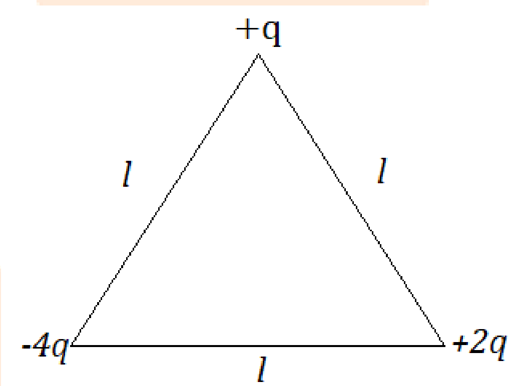
Derive the formula for the focal length of a thin lens with the help of refraction formula of spherical surface. **5**

8. On which factors, does the capacitance of a parallel plate capacitor depend ? A parallel plate air capacitor is of 3p F capacitance. When a plate of dielectric constant k and thickness 3d/4 is placed between the plates of the capacitors the capacitance of the capacitor becomes 6p F. Find the

magnitude of k, where d Lis the distance between the plates of the capacitor. **5**

OR

what is Coulomb's law of electric force ? Three point charges + q, - 4q and +2q are placed at the vertices of an equilateral triangle of sides 1, as shown in the figure. Obtain the expression for the magnitude of the resultant electric force acting on the charge +q. Also find out the work done to separate the charges in order to move them up to infinity.



5

9. Write the Bohr's postulates for the hydrogen atom. Derive the formula for the orbital frequency of electron moving in the n th orbit of hydrogen atom, using Bohr's postulates. **5**

OR

What is meant by half-life of a radioactive element ? Obtain the formula of half-life with the help of

$N = N_o e^{-\lambda t}$. The activity of a certain radioactive element is R_1 at time t_1 and R_a at time t_2 Obtain the formula of average life of the element, where $(t_2 > t_1)$ **5**